Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 17: Approved Solutions

Equivalence Classes and Partitions

Covers Chartrand & Zhang, Section 5.2

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Compute one class, then find one you missed

Finding all the distinct classes

1. Pick any element a and compute $[a] = \{x : x \, R \, a\}$.
2. Pick an element *not* in any class found so far, and compute its class.
3. Repeat until every element is accounted for.

By Theorem 5.23 the classes you produce are disjoint and cover the set, so this terminates and misses nothing.

The two-way correspondence

Equivalence relation → partition: the distinct classes are the blocks.
Partition → equivalence relation: relate a and b when they share a block.
If you are given either one, you can produce the other.

Three mistakes to expect

1. **Listing $[a]$ for every a and calling them all distinct.** Different labels often name the same class.
2. **Producing overlapping classes.** If two of your classes share an element they are the same class — recheck the relation.
3. **Forgetting the diagonal** when reconstructing R from a partition. Every (a, a) must be there.

Part II: Practice Problems

Section A — Computing classes

1. Let $S = \{1, 2, 3, 4, 5, 6, 7\}$ and let R be the equivalence relation

$$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (7,7), (1,5), (5,1), (2,4), (4,2), (2,7), (7,2), (4,7), (7,4)\}.$$

Determine the distinct equivalence classes.

Approved solution

$[1] = \{1, 5\}$, and $[5] = \{1, 5\}$ as well.
$[2] = \{2, 4, 7\}$, and $[4] = [7] = \{2, 4, 7\}$.
$[3] = \{3\}$ and $[6] = \{6\}$.
Distinct classes: $\{1, 5\}$, $\{2, 4, 7\}$, $\{3\}$, $\{6\}$ — four in all.
Check: they are pairwise disjoint and their union is S, so they partition S as Theorem 5.23 requires. Note $\{2, 4, 7\}$ required all three of $(2, 4)$, $(2, 7)$, $(4, 7)$ and their reverses to be present — transitivity forces that.

2. For each equivalence relation on $\mathbb{Z}$, describe the distinct equivalence classes.

(a) $a \; R \; b$ if $a - b$ is divisible by 4.

(b) $a \; R \; b$ if $a^2 = b^2$.

(c) $a \; R \; b$ if $a = b$.

Approved solution

(a) Four classes, by remainder on division by 4:

$$[0] = \{\ldots, -4, 0, 4, 8, \ldots\}, \quad [1] = \{\ldots, -3, 1, 5, \ldots\}, \quad [2], \quad [3].$$

(b) Infinitely many classes: $[0] = \{0\}$, and $[n] = \{n, -n\}$ for every $n \neq 0$. Note $[0]$ is the only class with one element.
(c) Infinitely many classes, each a single element: $[a] = \{a\}$. Equality is the finest equivalence relation — it separates everything.

The two extremes are worth naming. Equality gives one class per element; the relation $a \; R \; b$ for *all* a, b gives a single class containing everything. Every other equivalence relation sits between them.

Section B — Going backwards

3. Let $S = \{a, b, c, d, e, f\}$. If the distinct equivalence classes of R are $\{a, d\}$, $\{b, f\}$, and $\{c, e\}$, write out R and state how many ordered pairs it contains.

Approved solution

R contains every pair within a block and nothing across blocks:

$$\begin{aligned} R = \{&(a,a), (a,d), (d,a), (d,d),\\ &(b,b), (b,f), (f,b), (f,f),\\ &(c,c), (c,e), (e,c), (e,e)\}. \end{aligned}$$

Each block of size 2 contributes $2^2 = 4$ pairs, and there are three blocks, so $|R| = 12$.
In general, a partition with blocks of sizes $k_1, k_2, \ldots$ gives $\sum k_i^2$ pairs.

4. An equivalence relation on $S = \{1, 2, 3, 4, 5, 6\}$ has exactly three distinct classes, and you are told $3 \in [4]$, $[2] \cap [6] = \emptyset$, and $1 \in [3]$. Determine the classes.

Approved solution

From $3 \in [4]$ we get $[3] = [4]$, and from $1 \in [3]$ we get $[1] = [3]$. So 1, 3, and 4 lie in one class: $\{1, 3, 4\}$. That leaves 2, 5, 6 to fill the remaining two classes. Since $[2] \cap [6] = \emptyset$, elements 2 and 6 are in *different* classes. With only two classes left and three elements, 5 must join one of them — and either choice gives exactly three classes.
Two consistent answers: $\{1, 3, 4\}, \{2, 5\}, \{6\}$ or $\{1, 3, 4\}, \{2\}, \{5, 6\}$.

The problem does not have a unique solution, and saying so is part of a complete answer. Students who produce one and stop have not checked whether the constraints pin things down — a habit worth correcting now.

Section C — The theorem

5. Explain, in two or three sentences each, which property of an equivalence relation guarantees:

(a) that no equivalence class is empty;

(b) that two classes sharing an element must be equal.

Approved solution

(a) Reflexivity. For any $a \in A$ we have $a \; R \; a$, so $a \in [a]$ by the definition of equivalence class. Hence every class contains at least the element that names it.
(b) Symmetry and transitivity together. Suppose $x \in [a]$ and $x \in [b]$, so $x \; R \; a$ and $x \; R \; b$. By symmetry $a \; R \; x$, and combining with $x \; R \; b$ by transitivity gives $a \; R \; b$. By Theorem 5.22 this forces $[a] = [b]$.

Neither property alone suffices for (b): symmetry lets you reverse a link and transitivity lets you chain two, and you need both moves to get from a to b through x.

Section D — Writing

6. In five to eight complete sentences, explain why an equivalence relation and a partition are two descriptions of the same thing. Give a concrete example from computing, and say what plays the role of each.

Approved solution

Model answer. Given an equivalence relation on a set, its distinct equivalence classes are nonempty, cover the set, and never partly overlap, so they form a partition — that is Theorem 5.23. Going the other way, given a partition one can define a relation by declaring two elements related when they lie in the same block, and that relation is automatically reflexive, symmetric, and transitive because "same block" has all three properties. So each structure determines the other, and nothing is lost in

either translation. A concrete example is a set of files related when they share a file extension: the relation is "has the same extension as," and the partition is the grouping of files by type, with one block for each extension present. The relation tells you whether two particular files belong together; the partition shows you the whole grouping at once. They carry identical information in different shapes.

Full credit requires both directions of the correspondence, not just the observation that classes happen to partition. The example must identify what the relation is and what the blocks are.

Section E — Challenge (optional)

7. How many different equivalence relations are there on a set of three elements? List the corresponding partitions.

Approved solution

Five. By the correspondence, counting equivalence relations on $S = \{1, 2, 3\}$ is the same as counting partitions of S:

1. $\{1\}, \{2\}, \{3\}$ — the relation is equality.
2. $\{1, 2\}, \{3\}$
3. $\{1, 3\}, \{2\}$
4. $\{2, 3\}, \{1\}$
5. $\{1, 2, 3\}$ — everything related to everything.

Compare with Practice Set 15, Problem 5: there are $2^9 = 512$ relations on a three-element set, and only 5 of them are equivalence relations. The three properties are a severe restriction.
The counts $1, 2, 5, 15, 52, \ldots$ for sets of size $1, 2, 3, 4, 5$ are the *Bell numbers*, and they grow faster than any polynomial. Counting them is a Chapter 8 problem.

Instructor notes

- **Problem 4 has two valid answers.** That is deliberate. Students who report one and stop have not asked whether the constraints determine the answer — a habit that costs them on exams.
- **Problem 2(c) and its remark** set up the two extremes: equality is the finest equivalence relation, all-related is the coarsest. Useful framing for partial orders later.
- **Problem 7 paired with Practice Set 15 Problem 5** is striking: 512 relations, 5 equivalence relations. Worth putting both numbers on the board.
- **Problem 3** gives the $\sum k_i^2$ count, which shows up again when students count pairs in Chapter 8.
- Reserved for Homework 7: Section 5.2 even-numbered exercises.